

Density and cardinality

[Back to Project mathematical routes](#)

Scope

A faithful separable representation bounds the carrier, while the density obstruction bounds every norm-dense subset from below. Together they give exact continuum norm density and a counterexample existence statement, keeping density, carrier cardinality and Hilbert dimension distinct.

2 direct Cards + 55 reused prerequisites = 57 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: Exact norm density of the fixed atomic algebra
- Key result: A Naimark counterexample of continuum norm density

[Read this route with prerequisites](#)

Detailed route conditions

The faithful separable representation gives the carrier bound $|T| \leq \mathfrak{c}$. Independently, the density obstruction gives $\mathfrak{c} \leq |D|$ for every norm-dense $D \subseteq T$. Taking $D = T$ proves that the norm density is exactly $\mathfrak{c} = 2^{\aleph_0}$ and supplies the continuum-density counterexample.

Carrier cardinality, norm-density character and Hilbert-space dimension remain separate notions. The existence statement allows a possibly nonunital algebra and does not include a faithful separable representation as one of its quantified requirements. Its chosen witness happens to be the fixed unital algebra T .

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

Level 0 (6 Cards)

Level 0

The selected GNS representation of a pure-state class

A choice of representative pure states, fixed literally at a root, gives one concrete GNS representation per equivalence class.

`MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations

[Open Declaration Card](#) [Read exact source](#)

Level 0

Exact projection links for one approximately inner automorphism

A point-norm approximation followed by close-projection conjugacy gives exact support identities for an arbitrary projection family.

`MathlibAnnex.CStarAlgebra.exists_shell_family_of_approximately_inner`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: Choosing the CAR shell family from proved homogeneity

[Open Declaration Card](#) [Read exact source](#)

Level 0

The CAR algebra as a completion of finite matrix stages

Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation.

`MathlibAnnex.CStarAlgebra.CAR.Limit`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The product-vector state on the completed CAR algebra · Approximately inner homogeneity of pure CAR states · A finite-row average centralizes its matrix stage · The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra · The completed CAR algebra is infinite-dimensional · A two-sided inner intertwining sequence for two states

[Open Declaration Card](#) [Read exact source](#)

Level 0

A fixed representative family of CAR shell data

One family records a transporting automorphism and exact shell links for each selected pure-state class, with an explicitly fixed root component.

`MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: Choosing the CAR shell family from proved homogeneity · The matching GNS fiber retains exactly its cyclic line · Shell matching fixes the trace of transported flags · An inequivalent GNS fiber has no residual common range · Reconstructing a represented unitary from its shells and residual corner · Assembling selected GNS cyclic subspaces into an isometric source representation

[Open Declaration Card](#) [Read exact source](#)

Level 0 The closed operator algebra generated by a representation and extra operators Places a represented algebra and a chosen family of bounded operators in one concrete closed algebra. MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget Immediate Card prerequisites: None in this selected scope Used by in this scope: An irreducible operator algebra constructed from projection shells · Reconstructing a represented unitary from its shells and residual corner Open Declaration Card Read exact source
Level 0 The normalized trace on a finite CAR stage The normalized matrix trace provides a positive tracial functional compatible with the CAR inclusions. MathlibAnnex.CStarAlgebra.CAR.stageTrace Immediate Card prerequisites: None in this selected scope Used by in this scope: Constructing the normalized trace on the completed CAR algebra Open Declaration Card Read exact source
Level 1 (9 Cards) Level 1 A two-sided inner intertwining sequence for two states An inner sequence records both forward and actual inverse convergence data before any limiting automorphism is constructed. MathlibAnnex.CStarAlgebra.CAR.HasInnerIntertwiningSequence Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Two-sided inner sequences imply pure-state homogeneity · Pure CAR states admit two-sided inner intertwining sequences Open Declaration Card Read exact source
Level 1 Constructing the normalized trace on the completed CAR algebra Carries compatible bounded matrix traces through the algebraic limit, normed union and completion. MathlibAnnex.CStarAlgebra.CAR.trace Immediate Card prerequisites: The normalized trace on a finite CAR stage · The CAR algebra as a completion of finite matrix stages Used by in this scope: Shell matching fixes the trace of transported flags · Extending the CAR trace to the fixed shell target Open Declaration Card Read exact source

Level 1 The product-vector state on the completed CAR algebra Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion. <code>MathlibAnnex.CStarAlgebra.CAR.rootState</code>
Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Purity of the completed root state
Open Declaration Card Read exact source
Level 1 The completed CAR algebra is infinite-dimensional Rules out finite dimension by retaining matrix subalgebras of unbounded dimension. <code>MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional</code>
Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: The shell-generated algebra is not an algebra of all compact operators · An irreducible CAR representation has no nonzero compact image
Open Declaration Card Read exact source
Level 1 An irreducible operator algebra constructed from projection shells Shell partial isometries with rank-one limiting defects can be completed to unitary links that join inequivalent irreducible fibers. <code>MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_atomicShellModel</code>
Immediate Card prerequisites: The closed operator algebra generated by a representation and extra operators Used by in this scope: The atomic-shell construction on selected pure-GNS fibers
Open Declaration Card Read exact source
Level 1 Distinct GNS classes have no unitary intertwiner The class index makes the selected pure-GNS family pairwise unitarily inequivalent. <code>MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation</code>
Immediate Card prerequisites: The selected GNS representation of a pure-state class Used by in this scope: An inequivalent GNS fiber has no residual common range · The atomic-shell construction on selected pure-GNS fibers · Assembling selected GNS cyclic subspaces into an isometric source representation
Open Declaration Card Read exact source

Level 1 The decreasing root projections in the CAR completion The distinguished stage corners give a nested sequence of projections supporting the root state. MathlibAnnex.CStarAlgebra.CAR.rootFlag Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Shell matching fixes the trace of transported flags · Reconstructing a represented unitary from its shells and residual corner Open Declaration Card Read exact source
Level 1 Approximately inner homogeneity of pure CAR states The homogeneity property asks for exact state transport by an automorphism that is locally approximable by inner automorphisms. MathlibAnnex.CStarAlgebra.CAR.PureStateHomogeneity Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Two-sided inner sequences imply pure-state homogeneity Open Declaration Card Read exact source
Level 1 A finite-row average centralizes its matrix stage Turns an arbitrary element of the completed CAR algebra into one commuting with a prescribed finite matrix stage. MathlibAnnex.CStarAlgebra.CAR.ofStage_commute_rowAverage Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Exact vector transport along an almost central unitary path · Approximating another vector state along a unitary path Open Declaration Card Read exact source

Level 2 (5 Cards)

Level 2

Root compression becomes scalar in norm

Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra.

`MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError`

Immediate Card prerequisites: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range · Simplicity of the completed CAR algebra · A common root vector realizes every selected state through the generators · Assembling selected GNS cyclic subspaces into an isometric source representation

[Open Declaration Card](#) [Read exact source](#)

Level 2

Purity of the completed root state

Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra.

`MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState`

Immediate Card prerequisites: The product-vector state on the completed CAR algebra

Used by in this scope: Choosing the CAR shell family from proved homogeneity · The matching GNS fiber retains exactly its cyclic line · The atomic direct sum of the selected CAR representations · An inequivalent GNS fiber has no residual common range

[Open Declaration Card](#) [Read exact source](#)

Level 2

The atomic-shell construction on selected pure-GNS fibers

Pure-state GNS data supplies the irreducible and inequivalent fibers required by the generic shell construction.

`MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_pureAtomicShellModel`

Immediate Card prerequisites: Distinct GNS classes have no unitary intertwiner · An irreducible operator algebra constructed from projection shells

Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra

[Open Declaration Card](#) [Read exact source](#)

Level 2 Shell matching fixes the trace of transported flags Recovers exact trace values from the initial and final supports of shell links. <code>MathlibAnnex.CStarAlgebra.CAR.trace_transportedFlag</code> Immediate Card prerequisites: The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra · A fixed representative family of CAR shell data Used by in this scope: Transported flags vanish on vectors generated by a trace vector Open Declaration Card Read exact source
Level 2 Two-sided inner sequences imply pure-state homogeneity The forward and inverse limits close the homogeneity argument once an intertwining sequence is supplied for every pure-state pair. <code>MathlibAnnex.CStarAlgebra.CAR.homogeneity_of_innerIntertwiningSequences</code> Immediate Card prerequisites: Approximately inner homogeneity of pure CAR states · A two-sided inner intertwining sequence for two states Used by in this scope: Pure-state homogeneity of the completed CAR algebra Open Declaration Card Read exact source
Level 3 (5 Cards) Level 3 The atomic direct sum of the selected CAR representations All selected pure-GNS fibers of the completed CAR algebra act together on one Hilbert direct sum. <code>MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation</code> Immediate Card prerequisites: The selected GNS representation of a pure-state class · Purity of the completed root state Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra · Reconstructing a represented unitary from its shells and residual corner · Assembling selected GNS cyclic subspaces into an isometric source representation Open Declaration Card Read exact source Level 3 Simplicity of the completed CAR algebra Every nonzero closed two-sided ideal of the CAR algebra contains the identity. <code>MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_limit</code> Immediate Card prerequisites: Root compression becomes scalar in norm Used by in this scope: An irreducible CAR representation has no nonzero compact image Open Declaration Card Read exact source

Level 3 The matching GNS fiber retains exactly its cyclic line Identifies the residual projection of a transported CAR flag in its matching pure-state representation. <code>MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_rankOne</code>
Immediate Card prerequisites: Root compression becomes scalar in norm · The selected GNS representation of a pure-state class · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line
Open Declaration Card Read exact source
Level 3 Transported flags vanish on vectors generated by a trace vector Turns decay of projection traces into norm convergence on each source-orbit vector. <code>MathlibAnnex.CStarAlgebra.CAR.tendsto_transportedFlag_orbit_zero</code>
Immediate Card prerequisites: Shell matching fixes the trace of transported flags Used by in this scope: The target represented on the CAR trace GNS space
Open Declaration Card Read exact source
Level 3 An inequivalent GNS fiber has no residual common range Uses compression and cyclic transport to exclude fixed vectors in every other chosen pure-state class. <code>MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_zero</code>
Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line
Open Declaration Card Read exact source

Level 4 (4 Cards)

Level 4

The atomic common range is one embedded GNS line

Assembles the matching and inequivalent fiber calculations in an arbitrary Hilbert direct sum.

`MathlibAnnex.CStarAlgebra.CAR.iInf_range_atomic_transportFlag_eq_span`

Immediate Card prerequisites: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range

Used by in this scope: Every irreducible representation is unitarily equivalent to the inclusion · Realizing a shell family by a faithful irreducible operator algebra

[Open Declaration Card](#) [Read exact source](#)

Level 4

Assembling selected GNS cyclic subspaces into an isometric source representation

Joins mutually orthogonal pure-state cyclic copies and controls the limiting flag projections on their sum.

`MathlibAnnex.CStarAlgebra.CAR.exists_selectedAtomicCyclicIsometry`

Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations · A fixed representative family of CAR shell data

Used by in this scope: The cyclic sum fills every irreducible target representation

[Open Declaration Card](#) [Read exact source](#)

Level 4

Reconstructing a represented unitary from its shells and residual corner

Separates a represented generator into a strong shell sum and the exact operator between its limiting fixed spaces.

`MathlibAnnex.CStarAlgebra.CAR.exists_targetShellReconstruction`

Immediate Card prerequisites: The atomic direct sum of the selected CAR representations · The decreasing root projections in the CAR completion · A fixed representative family of CAR shell data · The closed operator algebra generated by a representation and extra operators

Used by in this scope: The target represented on the CAR trace GNS space · An irreducible target representation has a surviving fixed space

[Open Declaration Card](#) [Read exact source](#)

Level 4

An irreducible CAR representation has no nonzero compact image

Simplicity and infinite dimensionality rule out compact operators coming from nonzero CAR elements.

`MathlibAnnex.CStarAlgebra.CAR.eq_zero_of_isCompactOperator_image`

Immediate Card prerequisites: Simplicity of the completed CAR algebra · The completed CAR algebra is infinite-dimensional

Used by in this scope: Lifting a corner involution to an ambient CAR unitary · Realizing a finite Gram matrix in a represented CAR corner

[Open Declaration Card](#) [Read exact source](#)

Level 5 (4 Cards)

Level 5

Lifting a corner involution to an ambient CAR unitary

A prescribed unitary involution on a represented root corner can be realized on finitely many vectors by one lifted corner exponential.

`MathlibAnnex.CStarAlgebra.CAR.exists_liftedCornerExponential_apply_eq_involution`

Immediate Card prerequisites: An irreducible CAR representation has no nonzero compact image

Used by in this scope: Exact vector transport along an almost central unitary path · Approximating another vector state along a unitary path

[Open Declaration Card](#) [Read exact source](#)

Level 5

Realizing a shell family by a faithful irreducible operator algebra

Constructs unitary links between pure-state summands and an irreducible algebra containing a faithful copy of CAR.

`MathlibAnnex.CStarAlgebra.CAR.exists_completedAtomicShellModel`

Immediate Card prerequisites: The atomic common range is one embedded GNS line · The atomic direct sum of the selected CAR representations · The atomic-shell construction on selected pure-GNS fibers

Used by in this scope: The ambient inclusion of the fixed shell-family target · The CAR source map into the fixed shell-family target · The C^* -algebra generated by the CAR representation and the shell unitaries

[Open Declaration Card](#) [Read exact source](#)

Level 5

An irreducible target representation has a surviving fixed space

Rules out simultaneous disappearance of all limiting CAR flags by passing reduction through the reconstructed generators.

`MathlibAnnex.CStarAlgebra.CAR.exists_nonzero_targetFixedSpace`

Immediate Card prerequisites: Reconstructing a represented unitary from its shells and residual corner

Used by in this scope: A common root vector realizes every selected state through the generators

[Open Declaration Card](#) [Read exact source](#)

Level 5

Realizing a finite Gram matrix in a represented CAR corner

The infinite-dimensional root corner contains an isometric copy of any finite family of vectors.

`MathlibAnnex.CStarAlgebra.CAR.exists_rootCornerFamily_with_gram`

Immediate Card prerequisites: An irreducible CAR representation has no nonzero compact image

Used by in this scope: Purifying a vector state on a finite CAR stage

[Open Declaration Card](#) [Read exact source](#)

Level 6 (5 Cards)

Level 6 A common root vector realizes every selected state through the generators Builds a compatible family of unit vectors from a surviving residual space and identifies their CAR vector states. MathlibAnnex.CStarAlgebra.CAR.exists_targetDefectVectorFamily
Immediate Card prerequisites: Root compression becomes scalar in norm · An irreducible target representation has a surviving fixed space Used by in this scope: The cyclic sum fills every irreducible target representation
Open Declaration Card Read exact source
Level 6 Purifying a vector state on a finite CAR stage Every finite-stage restriction of a unit vector state can be realized by a unit vector in any irreducible CAR representation. MathlibAnnex.CStarAlgebra.CAR.exists_unitVector_vectorFunctional_eq_on_stage
Immediate Card prerequisites: Realizing a finite Gram matrix in a represented CAR corner Used by in this scope: Approximating another vector state along a unitary path · Cross-representation state approximation with a protected finite set
Open Declaration Card Read exact source
Level 6 Exact vector transport along an almost central unitary path Finite matrix-state tests ensure exact transport while protecting a prescribed finite set throughout the path. MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_exact_unitary_path_apply_eq_and_conjugate_sub_norm_lt
Immediate Card prerequisites: Lifting a corner involution to an ambient CAR unitary · A finite-row average centralizes its matrix stage Used by in this scope: Cross-representation state approximation with a protected finite set
Open Declaration Card Read exact source

Level 6 The CAR source map into the fixed shell-family target Restricting the codomain of the selected atomic representation gives the source map into one fixed generated target. MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom Immediate Card prerequisites: Realizing a shell family by a faithful irreducible operator algebra Used by in this scope: The shell-generated algebra is not an algebra of all compact operators · Closed-ideal simplicity of the shell-generated algebra · Extending the CAR trace to the fixed shell target Open Declaration Card Read exact source
Level 6 The ambient inclusion of the fixed shell-family target The same concrete target has a literal representation on the selected atomic Hilbert space. MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion Immediate Card prerequisites: Realizing a shell family by a faithful irreducible operator algebra Used by in this scope: Closed-ideal simplicity of the shell-generated algebra Open Declaration Card Read exact source
Level 7 (5 Cards) Level 7 Approximating another vector state along a unitary path A fixed irreducible CAR representation can approximate any vector state on finitely many elements by moving a given unit vector. MathlibAnnex.CStarAlgebra.CAR.exists_unitary_crossRepresentation_path_approx Immediate Card prerequisites: Purifying a vector state on a finite CAR stage · Lifting a corner involution to an ambient CAR unitary · A finite-row average centralizes its matrix stage Used by in this scope: Pure CAR states admit two-sided inner intertwining sequences Open Declaration Card Read exact source
Level 7 The shell-generated algebra is not an algebra of all compact operators Excludes an injective representation of the infinite-dimensional unital algebra onto all compact operators. MathlibAnnex.CStarAlgebra.CAR.not_isCompactOperatorModel_shellFamilyTarget Immediate Card prerequisites: The CAR source map into the fixed shell-family target · The completed CAR algebra is infinite-dimensional Used by in this scope: A Naimark counterexample of continuum norm density Open Declaration Card Read exact source

Level 7 Cross-representation state approximation with a protected finite set Finite entrywise tests allow new state approximation while an entire unitary path nearly fixes earlier data. <code>MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_crossRepresentation_path_approx</code>
Immediate Card prerequisites: Purifying a vector state on a finite CAR stage · Exact vector transport along an almost central unitary path Used by in this scope: Pure CAR states admit two-sided inner intertwining sequences
Open Declaration Card Read exact source
Level 7 Extending the CAR trace to the fixed shell target The source trace first becomes a state on the existing target algebra. <code>MathlibAnnex.CStarAlgebra.CAR.exists_state_extension_trace</code>
Immediate Card prerequisites: The CAR source map into the fixed shell-family target · Constructing the normalized trace on the completed CAR algebra Used by in this scope: The GNS representation of the target trace extension
Open Declaration Card Read exact source
Level 7 The cyclic sum fills every irreducible target representation Uses shell reconstruction and residual lines to turn an isometric CAR intertwiner into a unitary equivalence on the source. <code>MathlibAnnex.CStarAlgebra.CAR.exists_surjective_selectedAtomicCyclicIsometry</code>
Immediate Card prerequisites: A common root vector realizes every selected state through the generators · Assembling selected GNS cyclic subspaces into an isometric source representation Used by in this scope: Every irreducible representation is unitarily equivalent to the inclusion
Open Declaration Card Read exact source

Level 8 (3 Cards)

Level 8 Every irreducible representation is unitarily equivalent to the inclusion Extends the unitary equivalence from the CAR source to all additional generators and the norm-closed target. MathlibAnnex.CStarAlgebra.CAR.ambientInclusion_unitaryEquivalent Immediate Card prerequisites: The atomic common range is one embedded GNS line · The cyclic sum fills every irreducible target representation Used by in this scope: Closed-ideal simplicity of the shell-generated algebra Open Declaration Card Read exact source
Level 8 Pure CAR states admit two-sided inner intertwining sequences Finite-set vector-state transport supplies explicit corrections; their summable conjugation bounds and state estimates yield the required two-sided sequence. MathlibAnnex.CStarAlgebra.CAR.hasInnerIntertwiningSequence_of_pure Immediate Card prerequisites: Approximating another vector state along a unitary path · Cross-representation state approximation with a protected finite set · A two-sided inner intertwining sequence for two states Used by in this scope: Pure-state homogeneity of the completed CAR algebra Open Declaration Card Read exact source
Level 8 The GNS representation of the target trace extension The chosen state on the existing target supplies its cyclic GNS representation. MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation Immediate Card prerequisites: Extending the CAR trace to the fixed shell target Used by in this scope: The target represented on the CAR trace GNS space · Faithfulness of the target-state GNS representation Open Declaration Card Read exact source

Level 9 (3 Cards)

Level 9 The target represented on the CAR trace GNS space A fixed pointed unitary transports the target action onto the Hilbert space constructed from CAR alone. MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation Immediate Card prerequisites: The GNS representation of the target trace extension · Transported flags vanish on vectors generated by a trace vector · Reconstructing a represented unitary from its shells and residual corner Used by in this scope: The fixed algebra on the CAR trace Hilbert space · Faithfulness of the representation on the CAR trace Hilbert space Open Declaration Card Read exact source
Level 9 Pure-state homogeneity of the completed CAR algebra The constructed inner intertwining sequences discharge the supplier hypothesis and give an approximately inner transporting automorphism. MathlibAnnex.CStarAlgebra.CAR.homogeneity Immediate Card prerequisites: Two-sided inner sequences imply pure-state homogeneity · Pure CAR states admit two-sided inner intertwining sequences Used by in this scope: Choosing the CAR shell family from proved homogeneity Open Declaration Card Read exact source
Level 9 Closed-ideal simplicity of the shell-generated algebra Uses a faithful irreducible representation in the unique unitary-equivalence class to rule out proper nonzero closed two-sided ideals. MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_shellFamilyTarget Immediate Card prerequisites: The ambient inclusion of the fixed shell-family target · Every irreducible representation is unitarily equivalent to the inclusion · The CAR source map into the fixed shell-family target Used by in this scope: Faithfulness of the target-state GNS representation Open Declaration Card Read exact source

Level 10 (2 Cards)

Level 10

Choosing the CAR shell family from proved homogeneity

CAR homogeneity and exact shell matching supply one family whose distinguished component is fixed by an explicit identity choice.

MathlibAnnex.CStarAlgebra.CAR.homogeneityShellFamily

Immediate Card prerequisites: Purity of the completed root state · Exact projection links for one approximately inner automorphism · A fixed representative family of CAR shell data · Pure-state homogeneity of the completed CAR algebra

Used by in this scope: The C*-algebra generated by the CAR representation and the shell unitaries

[Open Declaration Card](#) [Read exact source](#)

Level 10

Faithfulness of the target-state GNS representation

The ideal dichotomy of the same target forces the constructed nonzero representation to be injective.

MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation_injective

Immediate Card prerequisites: The GNS representation of the target trace extension · Closed-ideal simplicity of the shell-generated algebra

Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space

[Open Declaration Card](#) [Read exact source](#)

Level 11 (2 Cards)

Level 11

The C*-algebra generated by the CAR representation and the shell unitaries

Defines the single algebra $\mathcal{T}$ used in the irreducible and separable faithful representations.

MathlibAnnex.CStarAlgebra.CAR.AtomicCounterexampleAlgebra

Immediate Card prerequisites: Choosing the CAR shell family from proved homogeneity · Realizing a shell family by a faithful irreducible operator algebra

Used by in this scope: The fixed algebra on the CAR trace Hilbert space

[Open Declaration Card](#) [Read exact source](#)

Level 11

Faithfulness of the representation on the CAR trace Hilbert space

Unitary conjugation preserves injectivity of the target-state representation.

MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation_injective

Immediate Card prerequisites: The target represented on the CAR trace GNS space · Faithfulness of the target-state GNS representation

Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space

[Open Declaration Card](#) [Read exact source](#)

Level 12 (1 Card)

Level 12
The fixed algebra on the CAR trace Hilbert space
Defines a representation of the same algebra T on the GNS Hilbert space of the CAR trace.
<code>MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation</code>
Immediate Card prerequisites: The target represented on the CAR trace GNS space · The C^* -algebra generated by the CAR representation and the shell unitaries
Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space
Open Declaration Card Read exact source

Level 13 (1 Card)

Level 13
Faithfulness of the representation on the CAR trace Hilbert space
Proves that the representation $\tilde{\rho} : T \rightarrow B(H_\tau)$ is injective.
<code>MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation_injective</code>
Immediate Card prerequisites: The fixed algebra on the CAR trace Hilbert space · Faithfulness of the representation on the CAR trace Hilbert space
Used by in this scope: Exact norm density of the fixed atomic algebra
Open Declaration Card Read exact source

Level 14 (1 Card)

Level 14
Exact norm density of the fixed atomic algebra
Combines a cardinality bound from a faithful representation on a separable Hilbert space with a lower bound for every norm-dense subset.
<code>MathlibAnnex.CStarAlgebra.CAR.hasDensityCharacter_atomicCounterexampleAlgebra</code>
Immediate Card prerequisites: Faithfulness of the representation on the CAR trace Hilbert space
Used by in this scope: A Naimark counterexample of continuum norm density
Open Declaration Card Read exact source

Level 15 (1 Card)

Level 15 A Naimark counterexample of continuum norm density Gives a C^*-algebra of norm density c whose nonzero irreducible representations form one unitary-equivalence class, without an identification with the compact operators. MathlibAnnex.CStarAlgebra.CAR.existsNaimarkCounterexampleOfDensity_continuum
Immediate Card prerequisites: Exact norm density of the fixed atomic algebra · The shell-generated algebra is not an algebra of all compact operators Used by in this scope: None in this selected scope
Open Declaration Card Read exact source